Meeting 03 - update and planing

tag	project

Data Preparation Summary: Socio-Spatial Impacts of the 2010 Drought in Guizhou

Part 1: Population Data Preparation

1. County-Level Population Panel (2008–2012)

To analyze demographic responses to environmental shocks, the study first constructed a county-level population panel for Guizhou Province from **2008 to 2012**. The data were derived primarily from **Guizhou Statistical Yearbooks (2009–2013)**, supplemented by the **2010 National Population Census**.

Key Variables:

- `pop_total`: Total population at year-end (in 10,000 persons)
- `pop_agri`: Agricultural/rural population (urban cores set to 0)
- `pop_density`: Population density (persons per km²)
- `urban_core`: Boolean indicator for county-level urban core status
- `city_name`: Prefecture-level city affiliation

Missing data for certain counties and years were interpolated linearly where appropriate. Counties established after 2010, such as **Wanshan District**, were only retained for post-2011 records, with earlier values set to `NaN`.

Urban core counties—such as Guiyang and Kaili—were flagged automatically and assigned `pop_agri = 0` to reflect their non-agricultural structure.

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2. Migration Proxy Construction

Migration dynamics were inferred from year-over-year changes in population, adjusted for natural growth where data permitted.

Step 1: Crude Migration Rate

A baseline migration proxy was calculated as the annual population change rate:

$$mig\_rate_t = \frac{\texttt{pop\_total}_t - \texttt{pop\_total}_{t-1}}{\texttt{pop\_total}_{t-1}}$$

This metric reflects net in- or out-migration trends under the assumption that natural growth is stable or accounted for separately.

Step 2: Net Migration Estimate

Where birth and death data were available or imputable, the following estimate was used:

$$net\_migration_t = \Delta \mathtt{pop\_total}_t - (\mathtt{births}_t - \mathtt{deaths}_t)$$

In counties without census-reported births/deaths, the study applied **provincial-level crude birth and death rates** (from statistical yearbooks) to estimate natural population growth. Flags `births_source` and `deaths_source` were included to indicate whether values were **estimated** or **directly sourced**.

Output Fields:

- `net_migration`: Estimated number of net migrants per year
- `net_migration_per_1000`: Normalized to per 1,000 persons
- `births`, `deaths`: Raw values used in estimation
- `births_source`, `deaths_source`: Flags ("real" / "estimated")

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3. Data Output and Integration

The final dataset, titled:

> Guizhou_0812_panel_data

was produced as a balanced panel of 88 counties over five years. It is designed to be merged with environmental variables (NDVI, drought, rainfall) using `county_name` and `year` as keys. This integration enables causal analysis of how drought intensity correlates with population shifts.

Variable Definitions for County-Level Population & Migration Panel (2008–2012)

Variable	Description	Unit / Type	Notes
`county_name`	County name	String	88 counties in Guizhou Province
`year`	Calendar year	Integer	2008–2012

Variable	Description	Unit / Type	Notes
city_name	Prefecture-level city	String	Administrative affiliation
pop_total	Total year-end population	10,000 persons	Interpolated if missing
pop_agri	Rural / agricultural population	10,000 persons	Set to 0 for urban core counties
pop_density	Population per unit area	persons / km^2	Calculated or interpolated
urban_core	Urban core status	Boolean (True/False)	Based on predefined city districts
treated	Drought treatment dummy	0 / 1	To be added from drought classification
post	Post-drought dummy (year ≥ 2010)	0 / 1	Fixed across all counties
mig_rate	Annual population change rate (proxy for migration)	Percentage	Δ pop_total / pop_total(t-1)
net_migration	Estimated net migration (adjusted for natural growth)	Persons	May be NaN if data unavailable
net_migration_per_1000	Net migration normalized per 1,000 persons	Persons / 1,000	Used for comparability across counties
births	Estimated or real number of births	Persons	Census-based or provincial ratio applied
deaths	Estimated or real number of deaths	Persons	Same as above
births_source	Data source for births	String ("real"/"estimated")	Quality flag
deaths_source	Data source for deaths	String ("real"/"estimated")	Quality flag

Part 2: Migration Proxy Construction

1. Study Scope and Design

This research aims to examine the socio-spatial impacts of the 2010 extreme drought in Guizhou Province, China. The study focuses on:

- Agricultural land use change (proxied by NDVI)
- Drought exposure (based on SPEI index)
- Environmental moderators (rainfall, cropland ratio)

A county-level panel dataset (2008–2012) has been constructed to support difference-in-differences and heterogeneity analyses.

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2. Core Data Sources

Variable	Source	Spatial Unit	Temporal Resolution
NDVI (all/cropland)	MODIS MOD13Q1	County	Annual, Growing Season (Apr–Sep)
Drought (SPEI)	SPEIbase v2.10	Pixel aggregated to county	2010 only
Rainfall	CHIRPS daily	County	Annual, Apr–Sep
Cropland ratio	MODIS MCD12Q1 (2010)	County	Static

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3. Drought Variable Construction

To capture the 2010 drought event, the study used the 3-month SPEI (Standardized Precipitation Evapotranspiration Index) from the **SPEIbase v2.10** dataset. The selected time window was **February to June 2010**, corresponding to the peak of the prolonged drought period in Southwest China.

Each pixel was classified as experiencing drought if its SPEI value was less than -1.5. A binary drought mask (0 or 1) was generated for each month, and the average across the five months was computed to produce a **drought frequency image**, representing the proportion of months during which drought occurred.

This frequency image was then **aggregated at the county level** using the **median value of all pixels within each county**, based on `reduceRegions()` and `ee.Reducer.median()`, to capture the central drought tendency for each administrative unit.

To categorize drought severity, the study applied **quartile-based thresholds** to assign counties to one of four levels (`drought_class`):

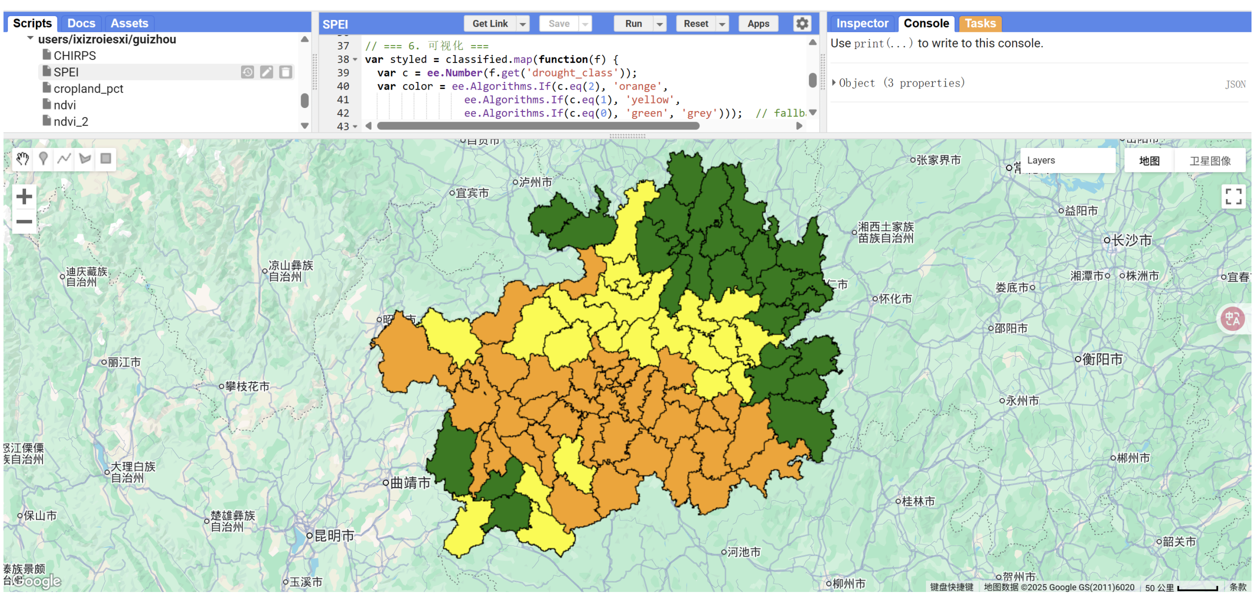
- 0: counties with drought frequency in the bottom 40%
- 1: 40%–60% percentile
- 2: 60%–80% percentile
- 3: top 20% of counties with the highest drought frequency

This classification reflects **relative drought intensity** across space rather than an absolute scale.

The final outputs include:

- `drought_pixel_rate`: mean pixel-level drought frequency (range: 0 to 1)
- `drought_class`: categorical drought severity (0–3)
- `median`: county-level median drought frequency

The results were visualized and exported in CSV format for integration with the NDVI and rainfall panel data.



```
// === 用户参数设置 ===
var counties = ee.FeatureCollection("projects/casa0004dissertation/assets/guizhou_county");

// === Step 1: 加载 SPEI 3-month 数据并选定时间段 ===
var spei03 = ee.ImageCollection("CSIC/SPEI/2_10")
  .filterDate('2010-02-01', '2010-06-30')
  .select('SPEI_03_month');

// === 2. 计算每张影像的干旱掩膜（SPEI < -1.5） ===
var droughtBinaryCollection = spei03.map(function(img) {
  return img.lt(-1.5).rename('drought');  // 返回0或1的图像
});

// === 3. 将5个月的干旱掩膜取平均，得到干旱频率（0-1） ===
var droughtFrequency = droughtBinaryCollection.mean();  // 每个像元干旱频率

// Step 4: 以中位数作为频率代表值（更稳健）
var droughtStats = droughtFrequency.reduceRegions({
  collection: counties,
  reducer: ee.Reducer.median(),
  scale: 10000  // 可选更高分辨率
});

// Step 5: 使用四分位数分类 drought_class（相对干旱强度）
var classified = droughtStats.map(function(f) {
  var freq = ee.Number(f.get('median'));
  var cls = ee.Algorithms.If(freq.lte(0.4), 0,
    ee.Algorithms.If(freq.lte(0.6), 1,
    ee.Algorithms.If(freq.lte(0.8), 2, 3)));
  return f.set({
    'drought_pixel_rate': freq,
    'drought_class': cls
  });
});
```

```
// === 6. 可视化 ===
var styled = classified.map(function(f) {
  var c = ee.Number(f.get('drought_class'));
  var color = ee.Algorithms.If(c.eq(2), 'orange',
              ee.Algorithms.If(c.eq(1), 'yellow',
              ee.Algorithms.If(c.eq(0), 'green', 'grey')));  // fallback 加 grey
  return f.set('style', {
    color: 'black',
    width: 1,
    fillColor: color,
    fillOpacity: 0.5
  });
});


Map.centerObject(counties, 7);  // 居中显示贵州
Map.addLayer(droughtFrequency.clip(counties.geometry()),
             {min: 0, max: 2, palette: ['green','orange','red']},
             'SPEI Drought Frequency (Feb–Jun 2010)');
Map.addLayer(styled.style({styleProperty: 'style'}), {}, 'County Drought Class
(Q-Based)');

print(classified.aggregate_histogram('drought_class'));


// step6: 导出表格
Export.table.toDrive({
  collection: classified,
  description: 'guizhou_drought_class_2010_final',
  fileFormat: 'CSV'
});
```

4. NDVI Variable Construction

MODIS MOD13Q1 NDVI was used to measure agricultural vegetation dynamics during the growing season (April–September) for each year from 2008 to 2012. NDVI values were scaled by a factor of 0.0001 to convert to reflectance units.

To isolate vegetation activity specifically on agricultural land, NDVI values were masked using a cropland layer from **MODIS MCD12Q1 (2010)**. The mask used **IGBP land cover classification type 12**, representing croplands. This fixed cropland mask was applied to all five years, under the assumption that major land use categories remained relatively stable across the study period.

For each year, NDVI images within the growing season window were filtered, masked, and averaged to produce a single mean NDVI image per year. These annual NDVI composites were then aggregated at the county level using `reduceRegions()` at a 250m scale.

The result is a county-level panel dataset (`ndvi_cropland`) capturing annual cropland vegetation conditions from 2008 to 2012.

An unmasked version (`ndvi_all`) was also generated using the same time window for robustness checks and sensitivity analysis.

```
// === 1. 输入 ===
var counties = ee.FeatureCollection("projects/casa0004dissertation/assets/guizhou_county");

// === 2. NDVI 数据集（MODIS MOD13Q1, scaled）===
var modisNDVI = ee.ImageCollection("MODIS/006/MOD13Q1")
  .select('NDVI')
  .map(function(img) {
    return img.multiply(0.0001).copyProperties(img, ['system:time_start']);
  });

// === 3. MCD12Q1 耕地掩膜（IGBP classification, value 12）===
var cropland_mask = ee.ImageCollection("MODIS/006/MCD12Q1")
  .filter(ee.Filter.calendarRange(2010, 2010, 'year'))  // 使用2010年作为代表年
  .first()
  .select('LC_Type1')
  .eq(12);  // cropland = 12 in IGBP scheme
```

```
// === 4. 设置年份 + growing season 时间范围 ===
var years = ee.List.sequence(2008, 2012);
var startMonth = 4;
var endMonth = 9;

// === 5. 主循环：每年 growing season NDVI mean（仅耕地）===
var annualNDVI_cropland = years.map(function(y) {
  var year = ee.Number(y);
  var start = ee.Date.fromYMD(year, startMonth, 1);
  var end = ee.Date.fromYMD(year, endMonth + 1, 1);

  var ndviSeason = modisNDVI.filterDate(start, end)
    .map(function(img) {
      return img.updateMask(cropland_mask);  // 应用耕地掩膜
    });

  var meanNDVI = ndviSeason.mean().rename('ndvi_cropland');

  var stats = meanNDVI.reduceRegions({
    collection: counties,
    reducer: ee.Reducer.mean(),
    scale: 250,
  }).map(function(f) {
    return f.set('year', year);
  });

  return stats;
});

// === 6. 合并并导出 ===
var ndvi_crop_panel = ee.FeatureCollection(annualNDVI_cropland).flatten();

Export.table.toDrive({
  collection: ndvi_crop_panel,
  description: 'guizhou_ndvi_cropland_panel_2008_2012',
```

```
  fileFormat: 'CSV'
}
```

---

5. Rainfall Variable Construction

Total rainfall during each growing season (Apr–Sep):

To account for climate variability, the study computed total seasonal rainfall using the **CHIRPS daily precipitation dataset**. The analysis focused on the **growing season**, defined as the period from **April to September** for each year between **2008 and 2012**.

Daily CHIRPS images were filtered by year and growing season months, then summed to calculate **cumulative rainfall (in millimeters)** for each season. These annual rainfall composites were then aggregated to the **county level** using the `reduceRegions()` function with the **mean reducer** and a spatial resolution of 5 km, consistent with CHIRPS native resolution.

The resulting variable, `rain_gs`, represents **total growing season rainfall per county per year**, and was exported as a panel dataset for use as a control variable in subsequent statistical models.

```
// === Step 1: 输入县界 ===
var counties = ee.FeatureCollection("projects/casa0004dissertation/assets/guizhou_county");

// === Step 2: CHIRPS 数据集（每日降水，单位mm）===
var chirps = ee.ImageCollection("UCSB-CHG/CHIRPS/DAILY")
.select("precipitation");

// === Step 3: 设置年份与 Growing Season 范围 ===
var years = ee.List.sequence(2008, 2012);
var startMonth = 4;
var endMonth = 9;

// === Step 4: 主循环，逐年计算 growing season 总降水 ===
var annualRainfall = years.map(function(y) {
```

```
var year = ee.Number(y);
var start = ee.Date.fromYMD(year, startMonth, 1);
var end = ee.Date.fromYMD(year, endMonth + 1, 1);  // 包含9月

var season = chirps.filterDate(start, end);

// 累加 Growing Season 总降水
var totalRain = season.sum().rename('rain_gs');

// 空间聚合至县域
var stats = totalRain.reduceRegions({
collection: counties,
reducer: ee.Reducer.mean(),
scale: 5000,  // CHIRPS 分辨率约 5km
}).map(function(f) {
return f.set('year', year);
});

return stats;
});

// === Step 5: 合并输出 ===
var rain_panel = ee.FeatureCollection(annualRainfall).flatten();

Export.table.toDrive({
collection: rain_panel,
description: 'guizhou_chirps_rainfall_panel_2008_2012',
fileFormat: 'CSV'
});
```

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6. Cropland Proportion Variable Construction

To capture spatial variation in land use structure, the study constructed a static variable representing the proportion of cropland within each county.

Land cover data were obtained from the **MODIS MCD12Q1 (2010)** product, using the **IGBP classification scheme**. Pixels classified as **cropland (class 12)** were identified and compared to all valid land pixels (excluding class 0, which indicates missing data).

For each county, the total number of cropland pixels was divided by the number of valid pixels to calculate the **cropland share**, expressed as a percentage (`cropland_pct`). This measure reflects the relative dominance of cropland in each county in 2010 and is used to analyze heterogeneity in drought and NDVI responses across counties with different agricultural land structures.

The variable is static and only constructed for the baseline year (2010), under the assumption that major land use patterns remained relatively stable during the study period.

```
// === Step 1: 输入县界 ===
var counties = ee.FeatureCollection("projects/casa0004dissertation/assets/guizhou_county");

// === Step 2: 加载 MCD12Q1（2010）地表覆盖数据 ===
var lc2010 = ee.ImageCollection("MODIS/006/MCD12Q1")
  .filter(ee.Filter.calendarRange(2010, 2010, 'year'))
  .first()
  .select('LC_Type1');

// === Step 3: 生成 cropland 掩膜（IGBP 类型 12）===
var cropland = lc2010.eq(12);  // 耕地像元
var valid = lc2010.neq(0);     // 有效像元（排除缺失）

// === Step 4: 统计每县 cropland 占比 ===
var stats = counties.map(function(f) {
  var area = cropland.reduceRegion({
    reducer: ee.Reducer.sum(),
    geometry: f.geometry(),
    scale: 500,
    maxPixels: 1e8
  }).get('LC_Type1');
```

```
  var total = valid.reduceRegion({
    reducer: ee.Reducer.sum(),
    geometry: f.geometry(),
    scale: 500,
    maxPixels: 1e8
  }).get('LC_Type1');

  var pct = ee.Number(area).divide(ee.Number(total)).multiply(100);
  return f.set('cropland_pct', pct);
});

// === Step 5: 导出 ===
Export.table.toDrive({
  collection: stats,
  description: 'guizhou_cropland_pct_2010',
  fileFormat: 'CSV'
});
```

The final dataset includes 88 counties and 5 years (2008–2012), combining:

- NDVI (all and cropland-specific)
- SPEI drought severity (2010 only)
- CHIRPS rainfall (growing season)
- Cropland share (2010)

This structure supports causal inference using spatial fixed effects, DiD, and heterogeneity analyses.

Part 3: Final Panel Dataset Construction

The study constructed a county-level balanced panel dataset combining environmental, land use, and demographic indicators from **2008 to 2012** across **88 counties** in Guizhou Province.

Key Components

Variable	Description	Source / Notes
`ENG_NAME`	County name (in English transliteration)	Standardized across datasets
`year`	Calendar year	2008–2012
`ndvi_all`	County-level mean NDVI (all land pixels)	MODIS MOD13Q1
`ndvi_crop`	Mean NDVI masked by cropland pixels	MODIS MOD13Q1 + MCD12Q1 mask
`rain_gs`	Total growing season precipitation	CHIRPS (April–September)
`drought_class`	Drought severity classification (0–3)	SPEI-based, 2010 only
`drought_pixel_rate`	Drought frequency (proportion of drought months)	SPEI-based, 2010 only
`median`	County median drought frequency	Used in drought classification
`cropland_pct`	Share of cropland in total valid land cover (2010)	MODIS MCD12Q1

Data Integration Strategy

- Environmental variables (`ndvi_all` , `ndvi_crop` , `rain_gs`) are time-varying and fully observed for each year.
- Drought variables (`drought_class` , `drought_pixel_rate` , `median`) are only defined for **2010**, as the drought is treated as a one-time shock.
- Land cover proportion (`cropland_pct`) is treated as static and derived from 2010 land use data.
- Missing values in `drought_*` for other years are retained as `NaN` , and models are restricted accordingly.

Panel Size

- **Total observations**: 88 counties × 5 years = **440 rows**
- All variables are aligned by `ENG_NAME` and `year` , enabling seamless merging with additional demographic or outcome data (e.g., migration proxies).

Intended Usage

This panel structure enables:

- Difference-in-differences or event-study models
- Fixed effects regression with spatial and temporal controls
- Interaction terms for heterogeneity analysis (e.g., `drought_class × cropland_pct`)

Dissertation Progress & Timeline (as of July 15th)

Current Progress

- **Data preparation completed**
  - County-level panel dataset (2008–2012) for 88 counties
  - Variables: NDVI (cropland/all), SPEI drought class (2010), CHIRPS rainfall, cropland ratio, population and migration proxies
- **Merging and variable cleaning finished**
  - Balanced panel constructed and aligned on `county_name` and `year`
- **Initial exploratory analysis completed**
  - Time-series comparison of NDVI and migration across drought levels

Next Steps & Timeline

Date	Tasks	Notes
July 15–21	Run main regression models (DiD, fixed effects)	Test NDVI and migration impact with baseline drought treatment
July 22–28	Add heterogeneity analysis	Include interaction terms (e.g. drought × cropland_pct)
July 29–Aug 4	Conduct robustness checks and visualisation refinements	Sensitivity to drought thresholds, land use assumptions
Aug 5–10	Draft Results and Discussion chapters	Focus on spatial disparity and mechanisms
Aug 11–15	Finalise visual outputs and polish methodology section	Maps, charts, appendices

Date	Tasks	Notes
Aug 16–20	Proofreading, references, ethics statement, final formatting	Ensure compliance with submission guidelines
Aug 21	Final submission prep	Submit on Aug 22 via Moodle